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Computer Network Protocols

Chapter Four

Routing Algorithms & Protocols

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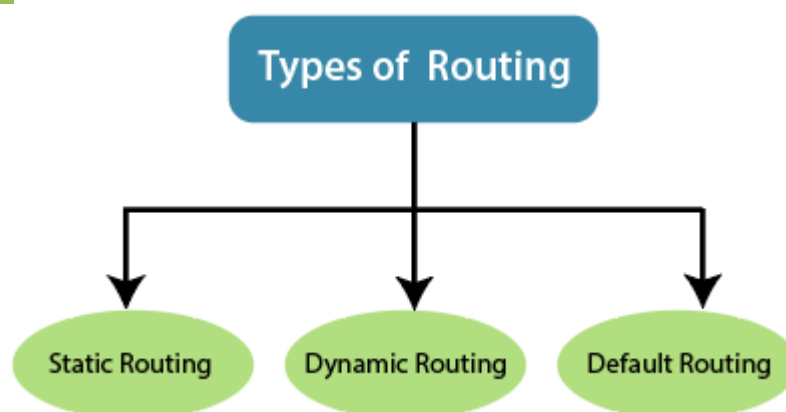
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4.1- Routing Concept & Algorithms

ROUTING: is **forwarding** of packets from one network to another network choosing the **best path** from the routing table.

Routing table consist of only the *best routes* for every destination.

Types of Routing:



Static	• It is configured by Administrator manually • Need for destination network ID • It is secure and fast • Used for small organization which have network of 10-15 routers
Dynamic	• Means automatically routing • Dynamic routes mean that the router learns the paths of destinations by receiving periodic updates from other routers • Is automatically choose the best shortest path • Can be done by using routing protocol

Q: list the Advantages of dynamic over static routing?

- There is **no need** to know the destination networks
- **Updates** the topology changes **dynamically**
- Administrative work **reduced**
- Used for **large organizations**

4.2- Routing Algorithms

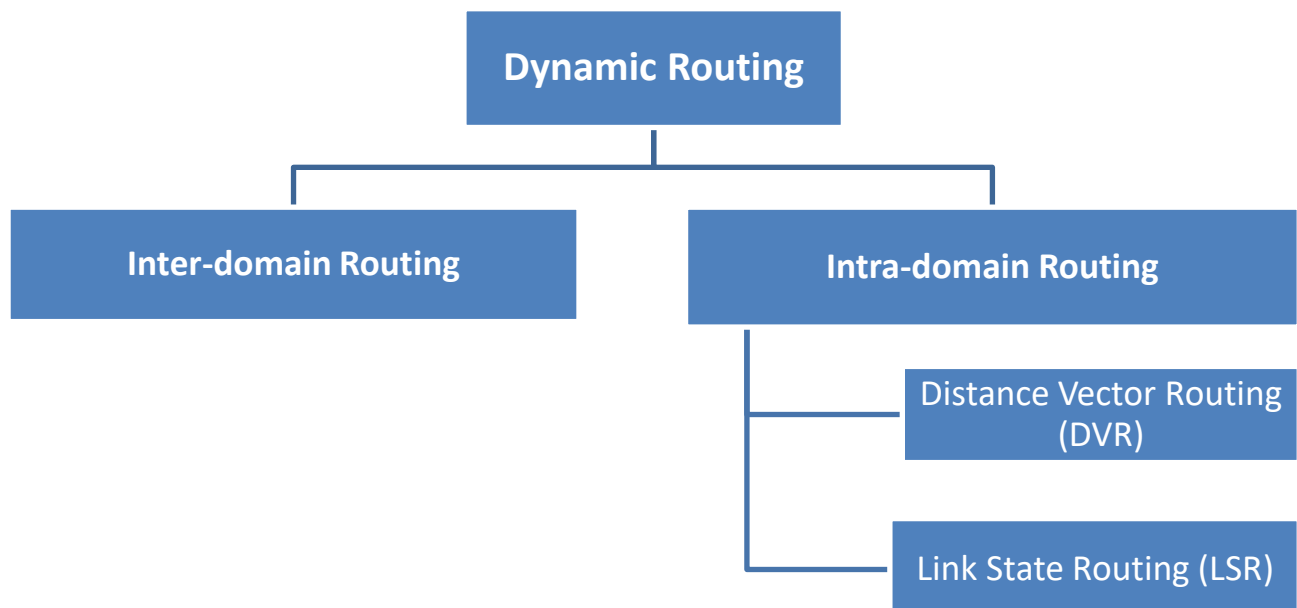
In the world of computers, any procedure can be illustrated through step by step, which is called an **algorithm**. Similarly, the routing algorithm is a step-by-step procedure for transferring data from one location to another over the network.

- The main function of the network layer is **routing packets from source to destination**.
- The **routing algorithm** is that part of the network layer software responsible for **deciding** which output line an incoming packet should be transmitted on.
- Routing algorithms can be grouped into two major classes: *non adaptive (Static)*, and *adaptive (Dynamic)*.

Dynamic Routing Algorithms

1- **Distance Vector Routing (DVR).**

2- **Link State Routing (LSR).**



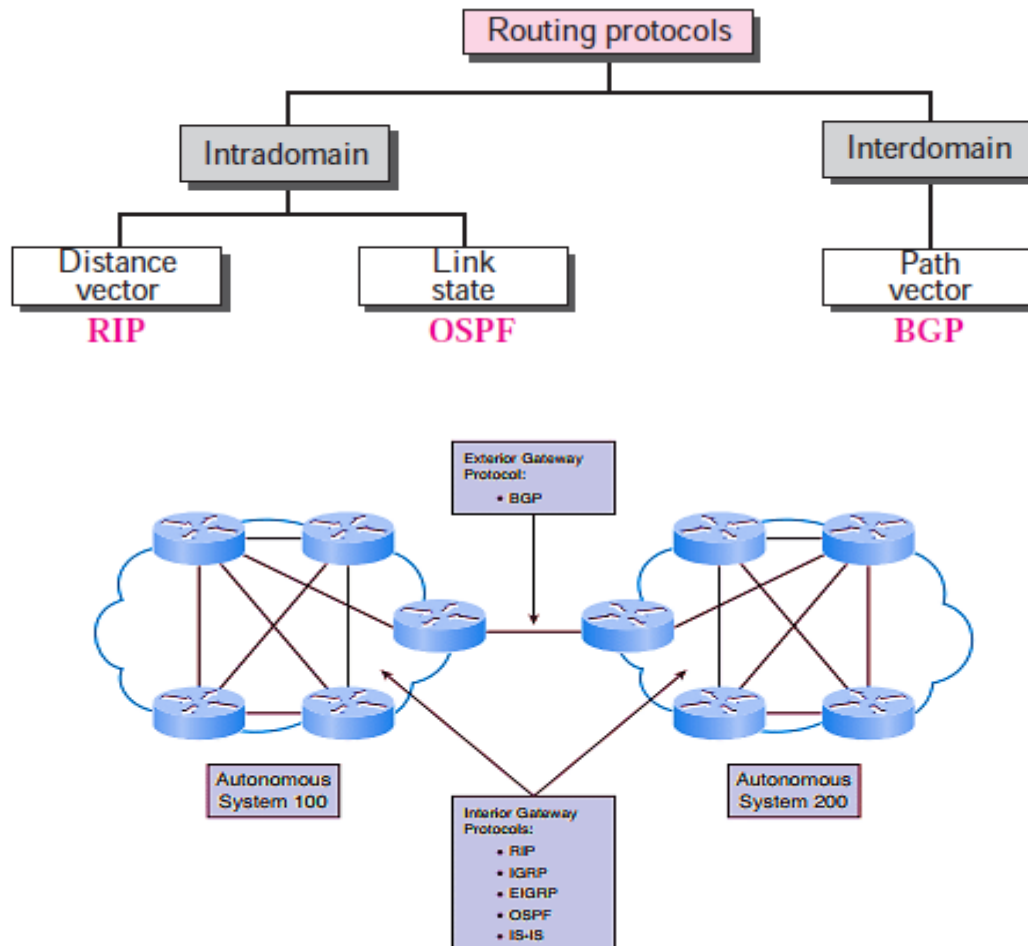
4.3- Routing Protocols

- A **routing protocol** is a combination of **rules and a procedure that lets** routers in the internet inform each other of **changes**.
- **Routing metric**: a method by which routing algorithms determines that one route is better than another route, Metric may be (*hop count, bandwidth, delay, or load*)

Interior Versus Exterior Routing Protocols

An autonomous system (AS), also known as a domain, is a **collection of routers** that are **under a common administration**, such as a company's internal network or an Internet service provider's (ISP's) network.

Because the Internet is based on the AS concept, two types of routing protocols are required:



4.4- Distance Vector Routing

- Distance Vector routing is **intra-domain** protocols, **inside** Autonomous system, but not between Autonomous system.
- distance-vector routing is based on the **least-cost** goal.
- Distance Vector developed by **Bellman-Ford** algorithm.
- Bellman **equation** is used to find the **least cost (shortest distance)** between a source to destination.
- A **distance vector routing algorithm** operates by having each router **maintain a table** (i.e., a **vector**) giving the best-known distance to each destination.
- These tables are **updated by exchanging** information with the neighbor's router. Every router knows the **best link** to reach each destination.
- **RIP** based on distance vector routing, each router **shares, at regular intervals**, its knowledge about entire AS with its neighbor.
- It is so **slow** and does not take **Bandwidth** into consideration when choose the root.

Distance vector algorithm

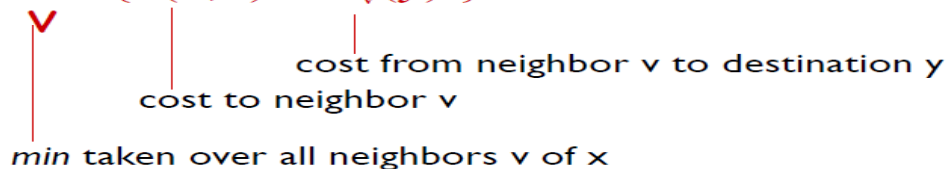
Bellman-Ford equation (dynamic programming)

let

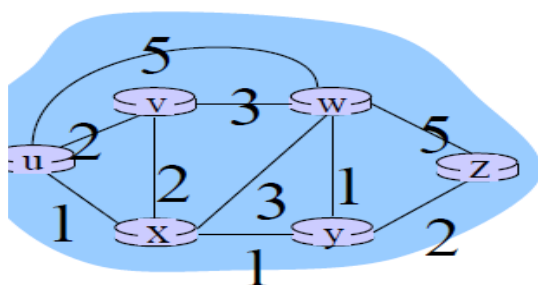
$d_x(y) := \text{cost of least-cost path from } x \text{ to } y$

Then

$$d_x(y) = \min \{ c(x,v) + d_v(y) \}$$



Bellman-Ford example



clearly, $d_v(z) = 5$, $d_x(z) = 3$, $d_w(z) = 3$

B-F equation says:

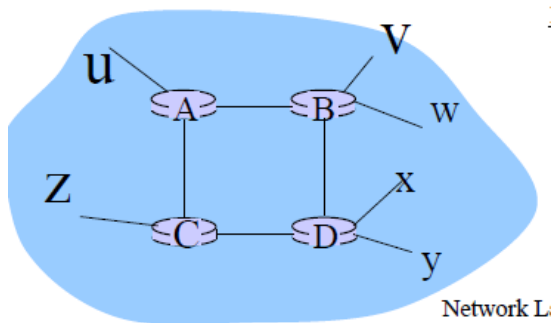
$$d_u(z) = \min \{ c(u,v) + d_v(z), \\ c(u,x) + d_x(z), \\ c(u,w) + d_w(z) \} \\ = \min \{ 2 + 5, \\ 1 + 3, \\ 5 + 3 \} = 4$$

node achieving minimum is next hop in shortest path, used in forwarding table

RIP (Routing Information Protocol)

- RIP is a routing protocol for exchanging routing table information between routers.
- It is a very simple protocol based on distance vector routing (DVR).
- simple intra-domain protocol
- Uses hop count as a path selection metric. (RIP prevents routing loops by implementing a limit on the number of hops allowed in a path from the source to a destination. *The maximum number of hops allowed for RIP is 15.* This hop limit, however, also limits the size of networks that RIP can support, a hop count of 16 is considered an **infinite distance**, in other words the route is considered unreachable.
- Three types of timers (**Routing-update timer (periodic timer) equal 30 seconds**, **Route timeout (Expiration Timer)**, if no update is received within the allotted 180 s, the route is considered expired and the hop count of the route is set to 16, which means the **destination is unreachable**, **Route-flush timer (Garbage Collection Timer)**, After the route timeout expires (240 s), the route-flush timer eventually expires, deleting the route from the table)

Example: Find router A routing table for the network topology shown below:



Dest. Net.	Next hop	Metric (Hop Count)
U	-	0
V	B	1
W	B	1
X	B	2
Y	B	2
Z	C	1

4.5- Link State Routing

The **primary problem in distance vector** that the algorithm often took **too long to converge** after the network topology changed (due to the count-to-infinity **problem**). Consequently, it was **replaced by** a new algorithm, now called link state routing.

The idea behind link state routing is simple and can be stated as five parts. Each router must:

- 1) **Discover** its neighbors and learn their network addresses.
- 2) **Measure** the delay or cost to each of its neighbors.
- 3) **Construct** a packet telling to all it has just learned.
- 4) **Send** the packet to all other routers.
- 5) **Compute** the shortest path to every other router.

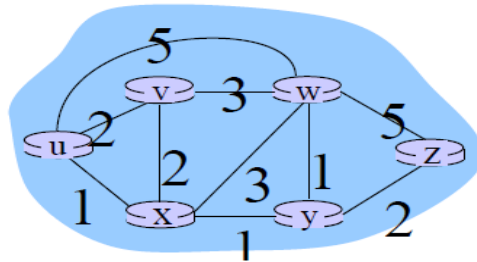
Compared to distance vector routing, link state routing requires **more memory and computation**. Also, the **computation time** grows faster

Nevertheless, in many practical situations, link state routing works **well because it does not suffer from slow convergence problems**.

Q: Consider the following network, define OSPF protocol & construct shortest path tree using Dijkstra algorithms

Dijkstra's algorithm: example

Step	N'	D(v),p(v)	D(w),p(w)	D(x),p(x)	D(y),p(y)	D(z),p(z)
0	u	2,u	5,u	1,u	∞	∞
1	ux	2,u	4,x		2,x	∞
2	uxy	2,u	3,y			4,y
3	uxyv		3,y			4,y
4	uxyvw					4,y
5	uxyvwz					



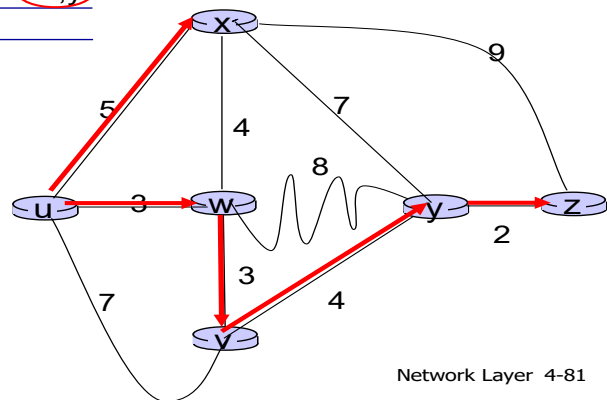
Q: Consider the following network, define OSPF protocol & construct shortest path tree using Dijkstra algorithms

Dijkstra's algorithm: example

Step	N'	D(v) p(v)	D(w) p(w)	D(x) p(x)	D(y) p(y)	D(z) p(z)
0	u	7,u	3,u	5,u	∞	∞
1	uw	6,w		5,u	11,w	∞
2	uwx	6,w			11,w	14,x
3	uwxv				10,y	14,x
4	uwxvy					12,y
5	uwxvyz					

notes:

- ❖ construct shortest path tree by tracing predecessor nodes
- ❖ ties can exist (can be broken arbitrarily)



Network Layer 4-81

Q: Compare between LSR & DVR

LSR	DVR
• All routers have complete topology, link cost info. • Updates are incremental and entire routing table is not sent as update. • Convergence is fast because of triggered updates. • More difficult to configure.	• Router knows physically-connected neighbors, link costs to neighbors • Entire routing table is sent as an update • Slow Convergence due to the count to infinity problem. • Easy to configure.

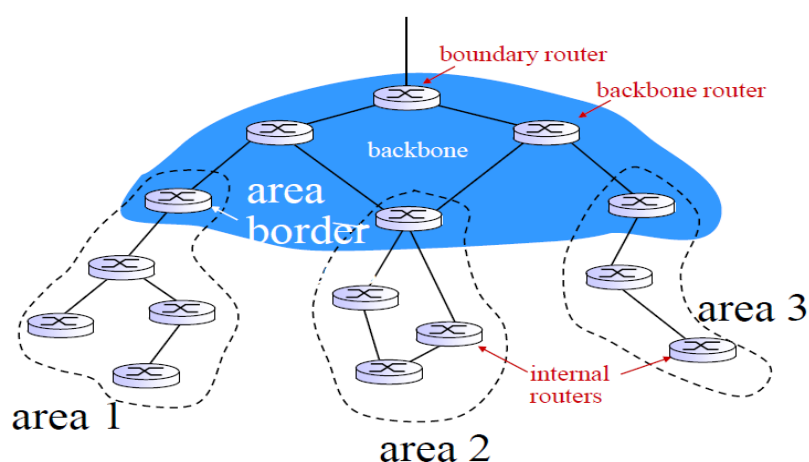
OSPF (Open Shortest Path First)

It uses a **link state routing algorithm** and falls into the group of interior routing protocols, operating within a single autonomous system (AS).

- **Topology map** at each node
- route computation using **Dijkstra's algorithm**
- OSPF is perhaps the most widely **used interior gateway protocol (IGP)** in large enterprise networks.
- The metric of OSPF is the **cost** of sending packets across a certain interface.
- **“open”**: publicly **available**
- **"Security"**: all OSPF messages authenticated (to **prevent malicious intrusion**)
- Hierarchical OSPF in large domains.

Hierarchical OSPF

- *Two-level hierarchy: local area, backbone.*
- Link-state advertisements only in **area**
- Each node has **detailed area topology**; only know direction (shortest path) to nets in other areas.
- **Area border routers**: “**summarize**” distances to nets in own area, advertise to other Area Border routers.
- **Backbone routers**: run OSPF routing limited to backbone.
- **Boundary routers**: connect to other AS's.



OSPF is superior to RIP in all aspects, including the following:

- It converges much faster.
- It supports hierarchical structures.
- It has improved metric calculation for best path selection.
- It does not have hop-count limitations
- At its inception, OSPF supported the largest networks.
- Compare to RIP, OSPF has no limitation due to hops (RIP has a limit of 15 hops so any network with more than 15 hops cannot be achieved by RIP).

4.6- Path Vector Routing

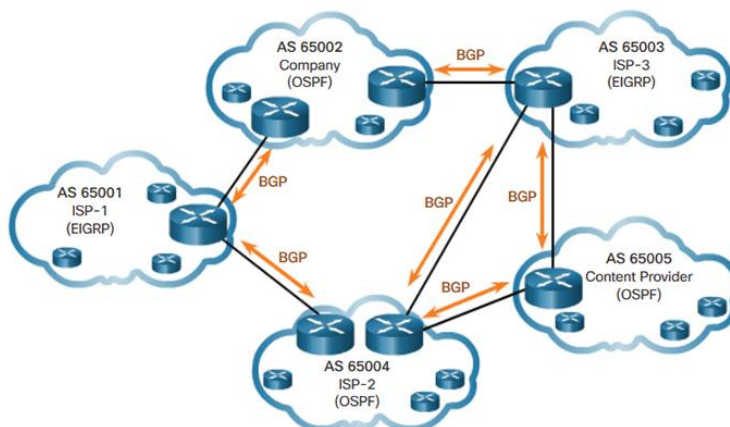
Path vector routing is an interdomain, policy-based protocol where routers advertise the entire path (a sequence of autonomous systems) to a destination, rather than just the distance/next-hop. It prevents routing loops by detecting if the local AS is already in the path, primarily used in the Border Gateway Protocol (BGP).

Key Aspects of Path Vector Routing

- **Loop Prevention:** Because the path vector contains the list of Autonomous Systems (AS) a packet will traverse, a router can instantly detect if a route loops back to its own AS and discard it.
- **Policy-Based:** Rather than just finding the shortest path, routers use policies (like cost, speed, or security) to select the best route, making it ideal for internet routing between different organizations.
- **Speaker Node:** In each AS, one specialized node ("speaker") advertises reaching network information to neighboring speaker nodes in other ASs.
- **Example (BGP):** The Border Gateway Protocol is the standard protocol for interdomain routing on the internet, using path vectors to ensure loop-free, policy-compliant path selection.

Comparison with Other Protocols

- **Vs. Distance Vector:** Distance vector only shares the next hop and distance (metric), leading to the "count to infinity" problem, whereas path vector eliminates this by listing the *full path*.
- **Vs. Link State:** Path vector is *more scalable* for large, interdomain networks and focuses on policy, while link state (like OSPF) provides faster convergence within a single domain.



Border Gateway Protocol (BGP)

- **Inter-domain routing protocol** for routing between autonomous systems (holds the Internet together)
- BGP is neither a link state, nor a distance vector protocol, it is known as **path vector protocol**. Routing messages in BGP contain complete routes.
- Network administrators can specify routing policies (**BGP supports flexibility** - paths could be chosen by a provider based on a policy).
- Network administrators can specify routing policies.
- BGP's goal is to find **any path** (not an optimal one).

Internal and external BGP sessions

iBGP	eBGP
• Used to connect different routers have same AS (same company) • Propagate reachability information to all AS-internal routers.	• Used to connect different routers have different AS (different company) • Obtain subnet reachability information from neighboring ASs.

